

Trainings ▾

Pressure Test

Monitor the rail fuel pressure. Use INSITE™ electronic service tool or connect a pressure gauge, Part Number 3824877, on the Compuchek™ fitting (as shown), and read the rail fuel pressure.

The rail pressure, while cranking at 130 rpm for at least 10 seconds, is 689 kPa [100 psi]. Make sure the fuel system is primed for at least 30 seconds.

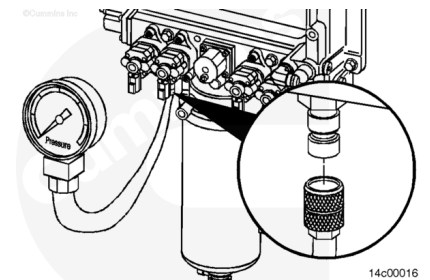
The rail fuel pressure with exhaust gas recirculation (EGR) **must** be 1585 to 1889 kPa [230 to 274 psi] at engine idle.

The rail fuel pressure with EGR **must** be 1896 to 2068 kPa [275 to 300 psi] at 2100 rpm.

The rail fuel pressure without EGR **must** be 1551 to 1889 kPa [225 to 274 psi] at engine idle.

The rail fuel pressure without EGR **must** be 1758 to 1999 kPa [255 to 290 psi] at 2100 rpm.

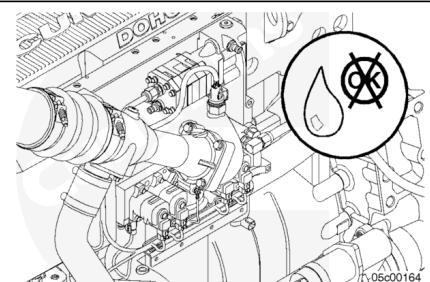
If the rail fuel pressure is below 689 kPa [100 psi], check to make sure there is fuel supply to the pump and that it is **not** restricted.



Leak Test

Clean the area surrounding the fuel pump. Run the engine at idle rpm.

Inspect for fuel leakage at the electronic control module (ECM) cooling plate supply and return lines and fittings. If a leak is detected, stop the test immediately and replace the damaged o-ring(s) or ECM cooling plate supply or return line(s).



Inspect for fuel or oil leakage at the fuel pump weep hole and integrated fuel system module (IFSM). If leaks are detected at the fuel pump weep hole or IFSM, stop the test immediately and replace the failed component.

Inspect for fuel or oil leakage at the fuel pump mounting gasket and o-rings. If leaks are detected in these areas, stop the test immediately and replace the damaged component.

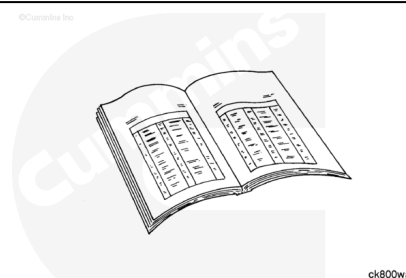
Inspect the head gasket for leaks.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries and cables. Refer to Procedure 013-009 in Section 13.
(</qs3/pubsys2/xml/en/procedures/10/10-013-009.html>)



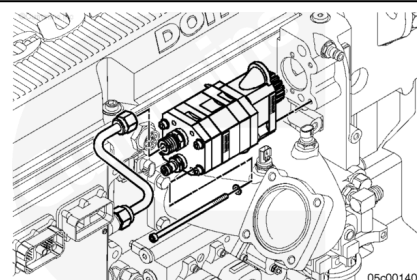
Remove

Automotive with CM870 and CM871

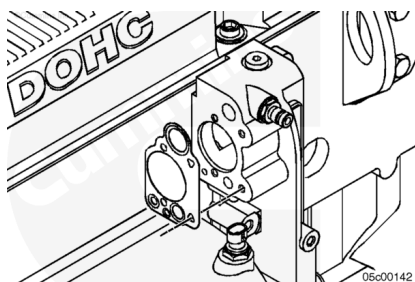
Clean the area surrounding the pump.

Disconnect the ECM cooling plate return line from the back of the fuel pump.

Remove the four fuel pump capscrews and remove the fuel pump from the IFSM.



Remove and discard the gasket.



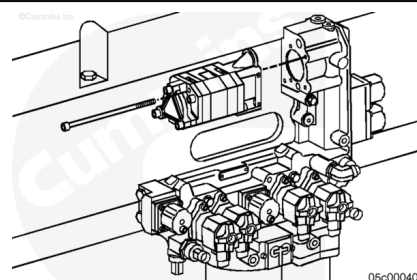
Automotive, Industrial with CM570 and Power Generation without CM570

Remove the fuel control cover.

Clean the area surrounding the pump.

Remove the four fuel pump capscrews and remove the fuel pump from the IFSM.

Discard the o-ring seals or gasket.

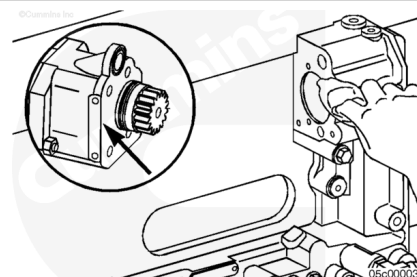


Clean and Inspect for Reuse

All Applications

Clean the gasket surfaces of the pump with a lint-free cloth.

Inspect the o-ring seal on the fuel gear pump. Replace the o-ring, if it is damaged.

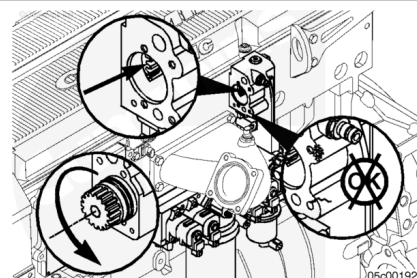


Inspect the mounting surfaces for cracks or damage. If cracks or damage are found, replace the fuel pump or IFSM.

Turn the fuel pump gear. If it spins without resistance or does **not** spin at all, replace the fuel pump.

Inspect the idler gear for wear or damage.

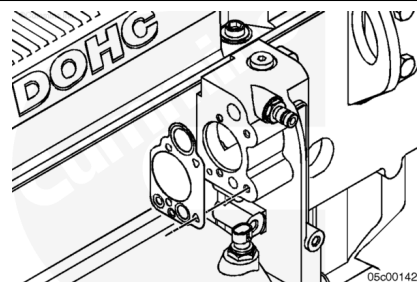
If the idler gear is worn or damaged, the IFSM **must** be replaced.



Install

Automotive with CM870 and CM871

Install a new gasket.



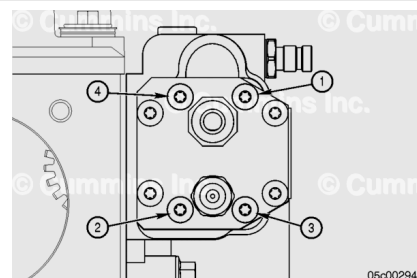
Install the pump onto the IFSM.

Install and tighten the four fuel pump capscrews in the sequence as shown in the illustration.

Torque Value: 6.5 n·m [58 in-lb]

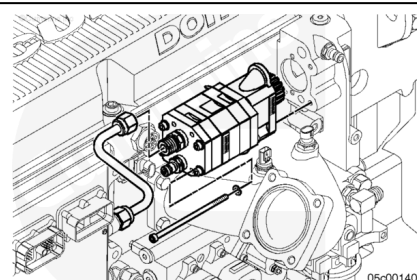
Tighten the four fuel pump capscrews in the sequence shown.

Torque Value: 13 n·m [115 in-lb]



Connect the ECM cooling plate return line to the back of the fuel pump.

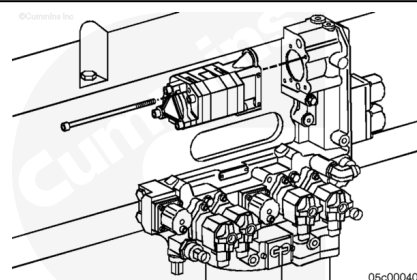
Torque Value: 25 n·m [221 in-lb]



Automotive, Industrial with CM570 and Power Generation without CM570

Replace the o-ring seals or gasket.

Install the pump onto the IFSM.



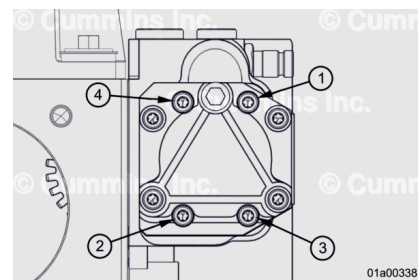
Install and tighten the four fuel pump capscrews in the sequence as shown in the illustration.

Torque Value: 7 n·m [62 in-lb]

Tighten the four fuel pump capscrews in sequence.

Torque Value: 13.5 n•m [120 in-lb]

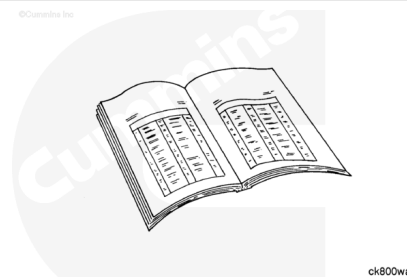
Install the fuel control cover.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Connect the batteries and cables. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/10/10-013-009.html)
- Operate the engine and check for leaks.

Prime

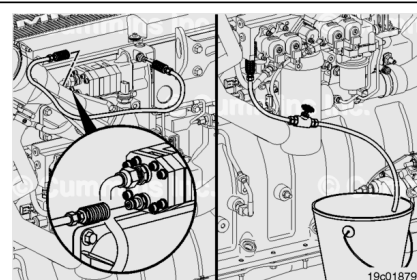
With Pressurized Fuel Filtering

To perform the priming procedure, use the following items to fabricate a fuel bypass hose:

- 2 - Quick-disconnect fittings, Part Number 3376859
- A length of clear tubing capable of 2758 kPa [400 psi].

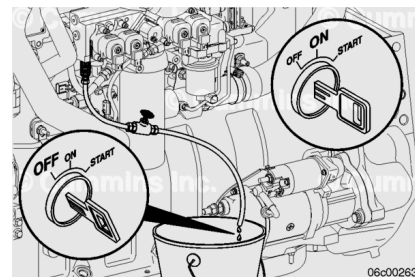
Install the fuel bypass hose fabricated above between the suction-side Compuchek™ fitting at the top of the IFSM and the fuel pump Compuchek™ fitting located on the head of the fuel pump.

Connect the clear hose and shutoff valve normally used to check for air in fuel to the rail Compuchek™ fitting. Refer to Procedure 006-003 in Section 6.



(/qs3/pubsys2/xml/en/procedures/10/10-006-003.html) Place the open end in a suitable container and fully open the shutoff valve in the hose.

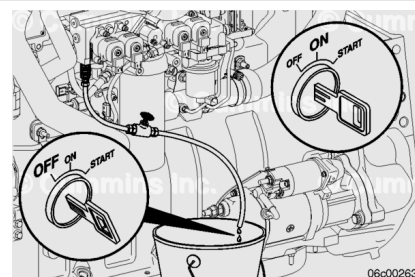
Fill the fuel filter by turning the ignition keyswitch to the ON position, but do **not** start the engine. The fuel lift pump should begin operating. Monitor the open end of the bleed line from the rail Compuchek™ fitting. When fuel comes out the end of this line, turn the ignition switch OFF. Remove the bleed line and air in fuel test line.



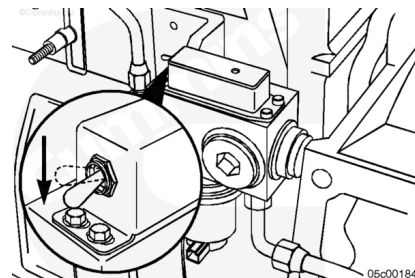
With Vacuum Fuel Filtering and Priming Pump

Note : It is possible the engine will operate rough for several minutes until all the air is out of the fuel system.

Fill the fuel filter by turning the ignition keyswitch to the ON position. The priming pump will operate for 2 minutes, which will adequately fill the fuel filter. The engine can then be started.



Some engines utilize a manual, remote mount priming pump. Flip the priming pump mounted toggle switch to the ON position. Operate the pump for 2 minutes and then turn it OFF. The engine can then be started.



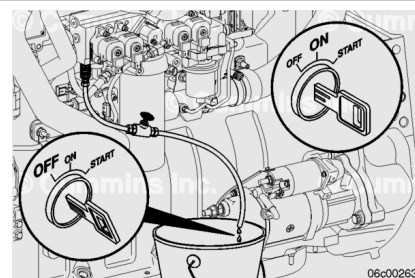
The manual switch priming pump can be activated by either the toggle switch on the priming pump or by cranking the engine

If the fuel system has lost prime, hold the switch on the pump to the ON position.

Operate the priming pump for 2 minutes, then turn it OFF.

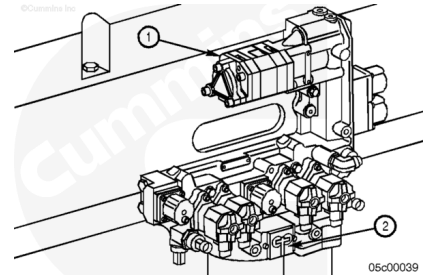
The engine can then be started.

It is possible the engine will operate rough for several minutes until all the air is out of the fuel system.



With Vacuum Fuel Filtering and Without Priming Pump

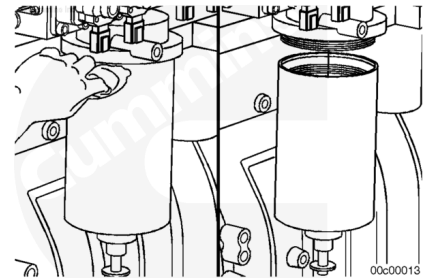
The Signature™, ISX, or QSX15 fuel system can require priming. If the engine is equipped with a priming pump, see the priming pump equipped Section above.



⚠ CAUTION ⚠

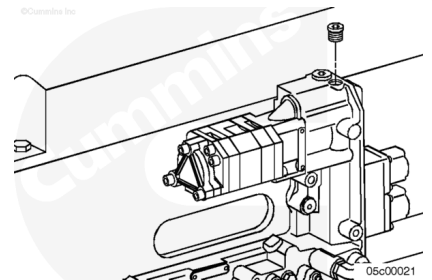
Fuel system damage can occur if debris is introduced into the clean side of the filter during the prefilling procedure.

Prefill the filter with a clean source of fuel.



Remove the top outboard M14 pressure side plug.

Note : Later versions of the IFSM have a 19-mm external hex plug at this location.



Crank the engine until fuel begins to flow from the top plug. Tighten the plug.

Crank the engine for 20 seconds. If the engine does **not** start within 20 seconds, wait 2 minutes. If necessary, repeat these steps until the engine starts.

It is possible the engine will operate rough for several minutes until all the air is out of the fuel system.

